BENCHMARKING KEYWORD SPOTTING ARCHITECTURES ACROSS TENSORFLOW LITE AND ONNX RUNTIME ON A RASPBERRY PI 5

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The Edge KWS Workload

- **Always-On Audio**: Continuously scans audio for wake-words.
- **Canonical Benchmark**: Standard task in MLPerf Tiny suite.
- **Strict Constraints**: Sub-millisecond latency (< 1 ms) and mW power.

Why On-Device? (SDG 3 Alignment)

- **Zero Cloud Dependency**: Works in offline rural clinics.
- **Strict Audio Privacy**: Biometric voice data never leaves device.
- **Target 3.8**: Hands-free assistive tech for patient diagnostics.

The Deployment Question

TinyML literature routinely assumes **TensorFlow Lite (TFLite)** is the fastest runtime on Arm. But on 64-bit Arm SBCs (Pi 5 Cortex-A76), **is TFLite universally faster than ONNX Runtime across diverse architectures?**

All models consume an identical $49 \times 10 \times 1$ **MFCC frame** and output a **12-way softmax**.

Model	Params	Structural Summary
DS-CNN (MLPerf Ref.)	22,604	Conv 10×4 (stride 2) + 4 $\times$ (DW 3×3 + PW 1×1) + GAP + Dense
DNN (Baseline)	88,396	4-layer MLP: 490–128–128–64–12 (lower-bound anchor)
LSTM (Compact)	19,980	Unrolled LSTM(64) over 49 time-steps + Dense head
CRNN (Hybrid)	56,844	Conv 3×3 ($\times 2$) + MaxPool ($\times 2$) + Unrolled GRU(64) + Dense

Decompiling Mixed-Precision MLPerf

The official MLPerf reference is actually **mixed-precision** (int8 conv, fp32 remainder). We decompiled and rebuilt a true-fp32 model for fair cross-runtime export.

Full INT8 Post-Training Quantization

Re-quantized with 1,000 representative clips (seed 42) into **true full-int8** flatbuffers (float I/O, int8 internal GEMM/conv operators).

EXPERIMENTAL RIGOR Controlled Benchmark Protocol on Raspberry Pi 5

Hardware & OS Environment

- **SoC**: Broadcom BCM2712 (Raspberry Pi 5)
- **CPU**: Quad-core Arm Cortex-A76 @ 2.4 GHz
- **RAM**: 8 GB LPDDR4X (synchronous)
- **OS**: Debian 64-bit GNU/Linux (Kernel 6.18)
- **Governor**: Pinned to `performance`
- **Runtimes**: LiteRT 2.1.6 vs ONNX Runtime 1.28

Statistical Protocol

- **Warmup**: 1,000 unmeasured consecutive passes
- **Sampling**: 1,000 sequential passes (never interleaved)
- **Outliers**: Filtered via $1.5 \times$ IQR rule
- **Bootstrapping**: 10,000 resamples for 95% CIs
- **Dual Test Sets (Speech Commands v2)**:
  - **10-Command Subset**: 4,074 official clips
  - **Full 12-Class Task**: 12,005 clips (+unknown/silence)

ACCURACY EVALUATION 10-Command Subset vs Full 12-Class Task

	10-Command Subset (4,074 clips)			Full 12-Class Task (12,005 clips)		
Model	**fp32 (%)**	**int8 (%)**	**Δ (pts)**	**fp32 (%)**	**int8 (%)**	**Δ (pts)**
DS-CNN (ref.)	90.55	90.57	**+0.02**	95.49	95.21	**−0.28**
DNN (anchor)	79.63	79.65	**+0.02**	72.65	71.90	**−0.75**
LSTM	90.48	87.43	**−3.05**	88.84	84.72	**−4.12**
CRNN	**92.56**	**92.17**	**−0.39**	**92.08**	**91.60**	**−0.48**

Numerical Parity

TFLite fp32 and ONNX fp32 match top-1 classifications across all 12,005 clips (diff $< 3 \times 10^{-6}$).

Recurrent PTQ Cost

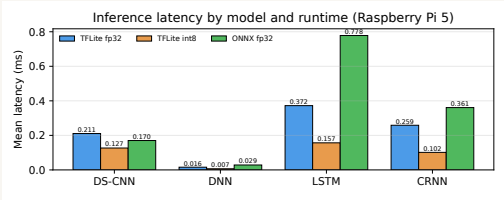
LSTM suffers large drop (**−4.12 pts** on full task), proving recurrence is hard to quantize without QAT.

CNN Resilience

DS-CNN and CRNN lose < 0.5 pts on the full task, demonstrating high robustness to full-int8 PTQ.

LATENCY BENCHMARK The Architecture Reversal Phenomenon

Model	TFLite	INT8	ONNX	Speedup
DS-CNN	0.2107	**0.1268**	**0.1702**	**0.81**× (ONNX)
DNN	**0.0158**	**0.0067**	0.0286	**1.81**× (TFLite)
LSTM	**0.3722**	**0.1569**	0.7782	**2.09**× (TFLite)
CRNN	**0.2586**	**0.1016**	0.3609	**1.40**× (TFLite)



Key Architectural Insight

- **Why ONNX wins on DS-CNN**: Operator-dispatch bound. ONNX fuses depthwise and pointwise convolution kernels.
- **Why TFLite wins on others**: GEMM and sequential bound. TFLite's lightweight session machinery wins by **1.40–2.09**×.

Full INT8 Acceleration

Full-int8 DS-CNN executes in **0.1268 ms** (25% faster than ONNX fp32 and mixed-precision artifact).

THREAD-SENSITIVITY SWEEP Deconfounding Multi-Thread Defaults (T = 1, 2, 4 Cores)

Model & Runtime	T=1	T=2	T=4
DS-CNN TFLite	0.2119	0.1229	**0.0782**
DS-CNN ONNX	0.4379	0.2594	0.1689
Ratio (ONNX / TFLite)	**2.07**×	**2.11**×	**2.16**×
CRNN TFLite	0.2598	**0.1873**	0.2195
CRNN ONNX	0.5075	0.4186	0.3610
Ratio (ONNX / TFLite)	**1.95**×	**2.24**×	**1.64**×

The Threading Confound

Default ONNX silently spawned a **4-thread pool**, while TFLite ran on **1 thread**!

Controlled Iso-Thread Results

- **TFLite Wins at Every Thread Count**: Pinned to $T = 1$, TFLite is **2.07**× **faster** than ONNX on DS-CNN (0.2119 vs 0.4379 ms).
- **Diminishing Returns on Small Models**: CRNN at 4 threads is slower than at 2 threads due to thread synchronization overhead.
- **Ranking Preserved**: The relative architecture ranking is completely preserved across core sweeps.

Model	Runtime	P_{Δ} (W)	t_{lat} (ms)	Energy E
DS-CNN	**TFLite fp32**	1.59	0.217	**0.34 mJ**
DS-CNN	ONNX fp32	1.18	0.448	0.53 mJ
CRNN	**TFLite fp32**	1.85	0.274	**0.51 mJ**
CRNN	ONNX fp32	1.58	0.494	0.78 mJ
LSTM	**TFLite fp32**	1.81	0.406	**0.74 mJ**
LSTM	ONNX fp32	1.31	0.725	0.95 mJ

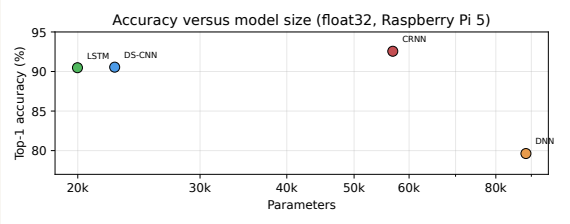
Direct On-Chip ADC Telemetry

Captured via `vcgencmd pmic_read_adc` across 12 voltage rails. Measures pure SoC energy, excluding peripherals.

The Power vs Energy Paradox

- **Peak Power**: TFLite draws higher instantaneous power ($+35\%$ on DS-CNN: 1.59 W vs 1.18 W).
- **Total Energy**: Because TFLite finishes $> 2\times$ faster, it consumes **35% less energy** per inference (0.34 mJ vs 0.53 mJ)!
- **Race-to-Sleep**: Completing bursts faster to return to idle saves substantially more battery life.

ACCURACY VS PARAMETERS FRONTIER Pareto Trade-offs on Speech Commands v2



Architecture Selection

- **CRNN (Top Accuracy)**: Highest accuracy (**92.56%** subset, **92.08%** full task) at 56.8k params.
- **LSTM (Parameter-Efficient)**: Reaches 90.48% with only **19,980 parameters**, but higher PTQ drop.
- **DS-CNN (Sweet Spot)**: Optimal balance: 22.6k params, 95.49% full task, 0.1268 ms latency.
- **DNN (Baseline Anchor)**: Spends 88.4k params for lowest accuracy (72.65% on 12-class).

CONCLUSION: PRACTICAL GUIDELINES Takeaways for Edge AI Practitioners

Core Findings

1. **Runtime Gap Does Not Generalize**: Defaulting to ONNX Runtime only benefits multi-threaded convolution; for dense and recurrent models, **TensorFlow Lite is 1.40–2.09× faster**.
2. **Iso-Thread Parity Favors TFLite**: When pinned to equal thread counts ($T = 1, 2, 4$), TensorFlow Lite is faster across all architectures.
3. **Quantization Exchange Rate**: Full-int8 yields a 1.7–2.5× speedup at negligible accuracy cost, except on recurrent units (LSTM).
4. **Race-to-Sleep Advantage**: Lower latency directly saves 35% battery energy per inference.

Actionable Recommendation

Choose your edge runtime based on architectural operator distribution, not convention.
Test both TFLite and ONNX with pinned threads before production deployment.

Thank You for Your Attention!

Questions & Discussion Welcome

Open Science Artifacts

- **Zenodo Permanent DOI**: `10.5281/zenodo.23765521`
- **GitHub Repository**: `github.com/jaweed3/kws-pi5-benchmark`
- **Open License**: GNU AGPLv3

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